

Econ 2 - Lecture 4 - 4/8/26

Lecture Quiz #2 Released Today, Due Monday, 4/13

Discussion Activity #1 this week, Activity #2 next week

Quarters are fast: Midterm Exam on Monday, April 27

Last class Recap

GDP: The market value of all final goods and services produced for a marketplace during a given period of time, within a country's borders

Convent Value of tracked production of non-intermediate, non-used goods/services that are newly produced.

GDP → geographical area
→ Gross State Product
→ Expand to continent, globe/world

Gross National Product (GNP): Production based on nationality

Sarah → Lives in UK → Part of UK GDP
↳ US citizen → Part of US GNP

How do we measure GDP?

Bureau of Economic Analysis (BEA)

Categorize GDP: Expenditure Approach

- 1.) Consumption (C): Households
- 2.) Investment (I): Firms
- 3.) Government Purchases (G): Government
- 4.) Net Exports (NX): foreign entities

$$GDP = Y = C + I + G + NX$$

Consumption (C) = ~67% of GDP

Goods: item we gain ownership of (tangible)

Durable: long-lasting goods (~3 yrs)

Non-Durable: frequently bought

→ ~45% of GDP ↳ food, gas, water bottle

Services: Intangible items we buy but do not gain ownership of

→ Repair → Haircut → Education

→ financial services → Subscriptions

→ Uber / Airline Tickets → Entertainment

Housing is $\sim 12\%$ of GDP

→ Rent an apartment → Service

→ Own an apartment unit? "Not a service?"

Imputed Rent

Estimate the value of every occupied dwelling on a monthly basis → add to GDP

10 homes on a street → all could rent for \$1,000/mo

Add 10 homes $\times$ \$1,000/home each month to GDP

Used home sales do not count!

New Home is only counted once

↳ Considered Investment, not consumption

Investment (I): Goods/Services bought by firms
($\sim 18-20\%$)

3 Subcategories

1. Non-Residential Investment = Capital

→ Tractor, machine, office furniture, software, plant/building, etc

2. Residential Investment: New Homes, Complexes

→ Non-moving, fixed asset

→ Production of "home goods"

3. Change in Private Inventory (+/- 0%)

Example: Inventory = Unsold goods

Campus Bookstore in 2024: \$10,000 in Hoodies

2024: GDP increased by \$10,000 $\rightarrow$ 8K in C
 $\rightarrow$ 2K in I

Imagine only \$8000 in hoodies are bought in 2024

Add \$8000 to Consumption

Remaining \$2000 $\rightarrow$ unsold product, added to inventory

Add \$2000 to investment in 2024

In 2025 $\rightarrow$ sell the leftover \$2000 in hoodies

Consumption $\uparrow$ by \$2000

Overall GDP in 2025 $\rightarrow$ no change $\rightarrow$ C $\uparrow$ 2000
 $\rightarrow$ I $\downarrow$ 2000

Inventory $\downarrow$ by \$2000, Investment $\downarrow$ by \$2000

2024: \$1 mil in watches produced

\$600k to households $\rightarrow$ C

\$200k to Switzerland $\rightarrow$ Exports

I? GDP?

\$200k leftover $\rightarrow$ unsold inventory $\rightarrow$ $\uparrow$ in I

Change in GDP = 1 mil = 600k + 200k + 200k
C X I

Government Purchase (G)

(~1790)

Goods/services bought by government

Necessities that are difficult to profit from

→ Infrastructure / roads, education

→ Parks, Police, Fire, National Defense

Note: Transfer Payments = Welfare, Social Security, Unemployment checks, etc
Do not count

Net Exports (NX) =

Exports
(X)

→ Money
is coming
in

→ Goods
leaving

- Imports
(M)

→ Money
is leaving

→ Goods
entering

$$\begin{array}{ccccccc} Y & = & C & + & I & + & G & + & NX \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \swarrow \searrow \\ 31.1T & & 21.11T & & 5.42T & & 5.32T & & 3.36T & 4.12T \\ & & & & & & & & X & M \\ & & & & & & & & \searrow \swarrow & \\ & & & & & & & & NX = & \\ & & & & & & & & -0.76T & \end{array}$$

Production

Alternative measures of GDP

GDP captures more than production

↳ Add up the value created of every intermediate good → GDP

Best Buy Computer example: Sum the firm profits
⇒ final value of \$400

Value - Added GDP = "Firm Profit"

When a firm makes profit, where does it go?

Best Buy sells a computer for \$400, \$50 in profit

→ Workers
→ Shareholders / owners
→ Lendlords
→ Debt holders

} "Income"

$GDP = Y = \text{Production} = \text{Value - Added} = \text{Income}$

Calculate GDP

UCSB Economy

<u>Products</u>	2009		2016	
	<u>Q</u>	<u>P</u>	<u>Q</u>	<u>P</u>
Food	1000	1	1500	2
Housing	100	100	150	150
Movies	500	5	1000	10

$$GDP_{2009} = Q_{09}^F \times P_{09}^F + Q_{09}^H \times P_{09}^H + Q_{09}^M \times P_{09}^M$$

$$\text{Nominal } GDP_{2009} = 1000 \times 1 + 100 \times 100 + 500 \times 5 = 13,500$$

↳ amount of dollars transacted in 2009

$$\text{Nom. } GDP_{09} = Q_{09} \times P_{09}$$

$$\text{Nominal } GDP_{2016} = Q_{2016} \times P_{2016}$$

$$= 1500 \times 2 + 150 \times 150 + 1000 \times 10$$

$$= 3000 + 22500 + 10000$$

$$= 35,500 \rightarrow \text{Value of all transaction in 2016}$$

Does nominal GDP represent more production?

→ Increase in Q & P

→ Hold prices constant on single base year

→ Real GDP

→ Hold prices constant at 2009 values

→ Real GDP in 2016?

$$\text{Real GDP}_{2016}^{\text{BY}=2009} = Q_{2016} \times P_{\text{BY}}, \text{ multiply \& sum}$$